

Effect of Traditional, Rest Redistribution, and Velocity-Based Prescription on Repeated Sprint Training Performance and Responses in Semiprofessional Athletes

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Abstract

Weakley, J, Castilla, AP, Ramos, AG, Banyard, H, Thurlow, F, Edwards, T, Morrison, M, McMahon, E, and Owen, C. The effect of traditional, rest redistribution, and velocity-based prescription on repeated sprint training performance and responses in semi-professional athletes. *J Strength Cond Res* XX(X): 000–000, 2022—The aim of this study was to investigate the effects of traditional, rest redistribution, and velocity-based repeated sprint training methods on repeated sprint performance, perceived effort, heart rate, and changes in force-velocity-power (FVP) profiles in male semiprofessional athletes. In a randomized crossover design, a traditional (2 sets of 6 repetitions [TRAD]), 2 different rest redistribution (4 sets of 3 repetitions [RR4] and 12 sets of 1 repetition [RR12]), and a 5% velocity loss (VL5%) (12 repetitions, with sets terminated when a 5% reduction in mean velocity had occurred) condition were completed. Mean and peak velocity, mean heart rate, and differential ratings of perceived exertion (dRPE) were measured throughout each session, while horizontal FVP profiles were assessed pre-session and post-session. The RR4 and RR12 conditions allowed the greatest maintenance of velocity, while the RR4, RR12, and VL5% had a moderate, significantly greater mean heart rate than the traditional condition. Trivial, nonsignificant differences between all conditions were observed in dRPE of the legs and breathlessness and FVP profiles. These findings indicate that rest redistribution can allow for greater maintenance of sprint velocity and heart rate, without altering perceived effort during repeated sprint training. In addition, velocity-loss thresholds may be a feasible method of prescription if athletes have diverse physical qualities and reductions in sprint performance during repeated sprint training are undesirable. Practitioners should consider these outcomes when designing repeated sprint training sessions because the strategic use of these methods can alter sprint performance and internal load without changing perceptions of intensity.

Key Words: repeated sprint training, velocity, heart rate, rest redistribution, velocity-based training

Introduction

Repeated sprint training involves the completion of maximal efforts of exercise for 3–7 seconds, interspersed with short recovery periods (i.e., <60 seconds) (4). This method of training is often used by team sport athletes to enhance repeated sprint ability, which is widely considered an important physical characteristic for sporting success (30). Akin to other methods of training, repeated sprint training is commonly implemented with a given number of sets and repetitions (e.g., 2 sets of 6 repetitions) (5,11,17). A decrement in sprint performance is typically observed as repetitions are completed, which is associated with increased neuromuscular fatigue and tax on metabolic pathways (20). Furthermore, accompanying these reductions in sprint

performance are changes in mechanical properties (14), which are commonly assessed through horizontal force-velocity-power (FVP) profiling (7,9,15). However, similar to other forms of training, the manipulation of the structure of the training session (e.g., rest redistribution) (16) may alter performance outcomes as well as internal and perceptual responses to training (22,37).

One method of structuring exercise that acutely maintains mechanical performance is the implementation of rest redistribution (22,31,32). Rest redistribution involves the reallocation of recovery time so that small intraset recovery periods are introduced during exercise. This can assist in the replenishment of adenosine triphosphate and phosphocreatine (PCr) which are primary contributors to anaerobic metabolism (31,33). For example, a set of 6 repetitions may be split into 2 sets of 3 repetitions with a small rest period (e.g., 20–60 seconds) between sets (31–33). By providing these small periods of recovery within training sets, greater kinetic and kinematic outputs have been

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shown to occur during resistance exercise (23,32,33). Furthermore, with these smaller, more frequent sets, reductions in perceptions of effort can occur (12). However, little is known about the effects of rest redistribution sets outside of resistance training, and it has received sparse attention during repeated sprints training (31). In addition, rest redistribution does not account for individual differences in repeated work capacity, which is an important consideration when working with an array of different athletes.

Velocity-based training (VBT) is a contemporary method of training prescription that allows individual differences to be taken into account and ensures the maintenance of performance (2,34). Velocity-based training often uses relative velocity loss thresholds (e.g., 5% velocity loss) to guide exercise termination and recovery allotment (26,35,36). Although VBT is commonly used during resistance training because of repeated sprinting often being performed over a predefined distance, velocity loss thresholds have also been transferred to running (1). For example, training can be structured so that if an athlete completes a sprint that equates to a drop in mean velocity of 5% or more compared with their first sprint (and often best repetition (10)), a recovery period can be permitted. By organizing repeated sprint training in this way, individual differences in repeated sprint ability, and the ability to recover between maximal efforts, could be accounted for.

Although repeated sprint training is used to develop repeated sprint ability, training prescription that follows strict set and repetition schemes are often used. This can cause substantial reductions in mechanical performance, which may mitigate the quality of the training session. By implementing alternative methods of structuring repeated sprint training that redistributes rest or accounts for individual ability to tolerate repeated maximal efforts, different acute training outcomes may occur. Consequently, the aim of this study was to investigate the effects of traditional, rest redistribution, and velocity-based repeated sprint training methods on repeated sprint performance, perceived effort, heart rate, and changes in FVP profiles in semiprofessional athletes.

Methods

Experimental Approach to the Problem

Across a 3-week period, this randomized crossover study assessed the acute effects of 4, quasi time-matched, repeated sprint training structures on perceptual measures (differential rating of perceived exertion [dRPE] for the legs and breathlessness was recorded on the 3rd, 6th, 9th, and 12th repetition), heart rate (mean), repetition velocity (mean and maximum), and sprint FVP profiles (preintervention and postintervention). The 4 protocols were 2 sets of 6 repetitions (TRAD), 4 rest redistributed sets of 3 repetitions (RR4), 12 evenly rest redistributed sets of 1 repetition (RR12), and 12 repetitions with rest permitted when velocity reduced by $\geq 5\%$ of the first repetition (VL5%). In total, subjects attended 5 sessions (i.e., familiarization and 4 testing sessions). Figure 1 presents an outline of the sessions.

Subjects

In G*Power a priori analysis was conducted to estimate the sample size required for this study, with a significance criterion of $\alpha = 0.05$ and power of 0.80, it was estimated that the minimum sample size needed for this study was 12. Consequently, 12, male,

semiprofessional rugby union players (age: 20.1 ± 0.8 years; body mass: 90.7 ± 11.5 kg; height: 1.82 ± 0.05 m; 30–15 Intermittent Fitness Test: 18.3 ± 1.0) from a British University and Colleges Super Rugby squad volunteered to take part in this study. Testing took place during the rugby preseason. All subjects were injury-free for at least 3 months before this study and did not complete any strenuous exercise in the 48 hours before the testing occasions. In addition, in the 48 hours before testing all subjects refrained from alcohol and in the 12 hours preceding testing, subjects did not consume any caffeine. Subjects were informed of the study procedures and signed a written informed consent form before initiating this study. The study protocol adhered to the Declaration of Helsinki and was approved by Leeds Beckett University Institutional Review Board (application reference: 55,423).

Protocols

This study was conducted across a 3-week period that included 1 familiarization and 4 testing occasions for each subject. All testing was completed on an outdoor synthetic running track, with subjects required to have 48 hours of active rest before all testing, and scheduled a time of day that was maintained across all 4 testing occasions. Temperature was consistent ($\pm 3^\circ$ C) across testing occasions for each subject, with minimal influence of wind because of the track being within a built-up area. During the first visit (i.e., familiarization), subjects signed consent forms and were provided the opportunity to ask any questions. In addition, they were measured for height (Seca Alpha stadiometer, model 213, Hamburg, Germany), explained the dRPE scale, and were provided their time of testing for the 4 testing occasions. On the second to fifth visits, subjects arrived, had their body mass measured (Seca 877 scales, Hamburg, Germany), and wore the same footwear, were fitted with the same heart rate monitor (Polar T31-coded, Polar Electro, Vantaa, Finland), numbered vest, and global positioning unit (Catapult OptimEye S5, Catapult Innovations, Melbourne, Australia) across each testing occasion (6,24).

On testing occasions, subjects were informed of their repeated sprint condition for that day and completed a standardized 15-minute warm-up consisting of 5 minutes of jogging and 10 minutes of lower-limb dynamic stretching. As part of the warm-up, subjects also performed 3 progressive sprints of 40 m at 50, 70, and 90% of perceived maximal effort (18). After the warm-up, subjects completed a maximal 40-m sprint that began in a crouching, 2-point staggered stance (8). During the 40-m sprint, velocity-time data were collected by a Stalker Acceleration Testing System (ATS) II radar device (Stalker ATS II, Applied Concepts, Dallas, TX) sampling at 46.7 Hz positioned on a tripod 5 m behind the start line and at a height of 1 m above the ground to align with the subjects' center of mass (7). To limit the possible variability introduced by manual operation, the radar device was operated remotely through connection to a laptop running Stalker ATS software (version 5.0.2.1; Applied Concepts). The data file for each trial were then imported together with the subject's height, body mass, and environmental conditions into a custom-made Microsoft Excel spreadsheet that was used to derive all sprint times and FVP characteristics, consistent with the procedures and formulas developed by Samozino et al. (28). The variables of interest from the sprint FVP profile have been extensively detailed elsewhere (7,9,15,21,28). After the FVP profiling, subjects were given 5 minutes of passive recovery before starting their respective repeated sprint protocol.

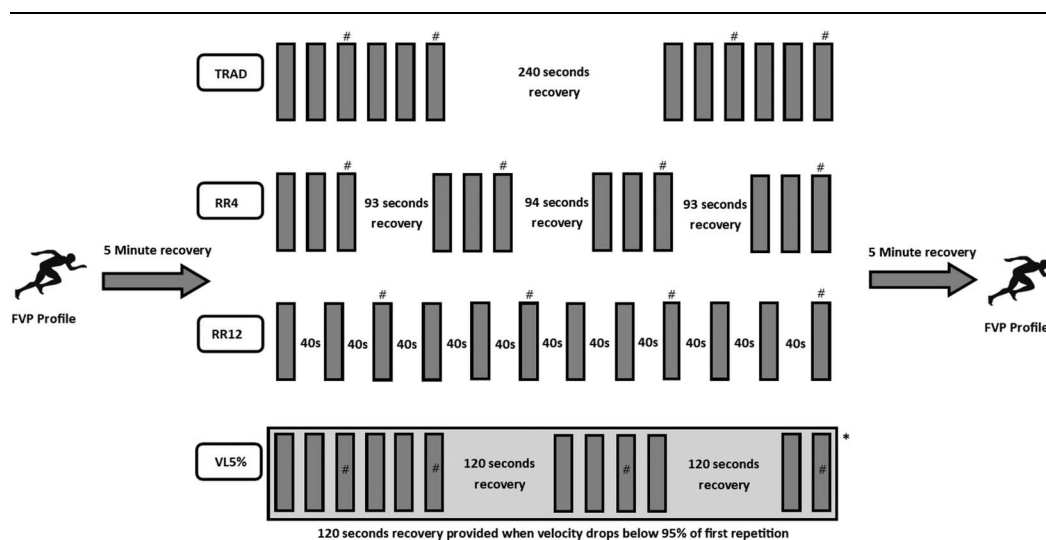


Figure 1. Outline of the 4 repeated sprint protocols. Recovery between repetitions was 20 seconds. Recovery between sets is outlined within the diagram in their respective condition. TRAD = traditional; RR4 = rest redistributed 4 sets of 3 repetitions; RR12 = rest redistributed 12 sets of 1 repetition; VL5% = velocity loss threshold of 5 percent; dRPE = differential rating of perceived exertion; * and gray background = theoretical example of VL5% condition; # = time point of dRPE measurement; FVP profile = force-velocity-power profile measurement.

During the 4 repeated sprint protocols, subjects were required to complete 12 repetitions of 30 m on a purposely made running track. Subjects started with their foot on a point 0.5 m behind the timing gate before each effort (39), with a 10-second warning and 3-second countdown provided for each repetition. Subjects were asked to use maximal intent for each effort, with loud verbal encouragement provided by the investigators where needed (38). During the recovery period between sprints, subjects decelerated and walked back to the starting point. During this study, 2 sets of timing gates (Brower Timing Systems, IREmit, Draper, UT) were used that worked in both directions. This allowed subjects to start each sprint at the end that they finished the previous sprint and reduced the requirement for subjects to rush back to a common starting point (20). Total rest (440 seconds) was equated across the TRAD, RR4, and RR12 conditions. Recovery between sets was 240 seconds in the TRAD, 93–94 seconds in the RR4, 40 seconds in the RR12, and 120 seconds in the VL5%. Recovery between repetitions was 20 seconds in the TRAD, RR4, and VL5% (note that the RR12 condition consisted of 12 sets of single sprints). Details of the sprint protocols are presented in Figure 1.

For the implementation of the VL5% condition, the time from the first sprint repetition was used to calculate the mean velocity/time threshold that would indicate if a loss of 5% mean velocity had occurred (e.g., if the first repetition speed was 4.40 seconds [i.e., mean velocity across 30 m of $6.82 \text{ m} \cdot \text{s}^{-1}$], then a 5% loss in mean velocity loss would be 4.63 seconds [i.e., $6.48 \text{ m} \cdot \text{s}^{-1}$]). A 5% velocity loss threshold was implemented based off extensive piloting which indicated that, on average, subjects would require 2 rest intervals. For this reason, the rest period between sets was standardized at 120 seconds which equated to the same period of rest that was provided in the other sprint protocols (i.e., between set recovery of 240 seconds).

The time of each repetition was recorded using timing gates, with this information being used to calculate mean velocity for each sprint. Global positioning unit, measuring at 10 Hz, was used for the calculation of peak velocity across each repetition (27). Similar to the method used by Tufano et al. (32) to assess the reduction in mean and peak velocity during each session, the

average of the final 3 repetitions was subtracted from the velocity of the first repetition and then divided by the velocity of the first repetition. This was then multiplied by 100 to calculate the decrement in performance across the session. Heart rate was recorded from the time of the initiation of the first sprint to exactly 1 minute after the 12th sprint. At the completion of the 3rd, 6th, 9th, and 12th repetition, dRPE for the legs and breathlessness was recorded on a CR100 scale (19,35). To collect dRPE, subjects were asked for their perception of breathlessness and leg muscle exertion and then required to state and point to the corresponding value on the CR100 scale (19). Furthermore, all testings were performed privately so that values could not be influenced by peers, and data were recorded manually by the lead investigator. Finally, after the completion of the final repetition, 5 minutes were provided to each subject. After this, the same testing procedure that was completed to record the FVP profile before the protocol was repeated.

Statistical Analysis

Hierarchical linear mixed models were used to identify the between protocol mean heart rate (HR) and dRPE and within condition change in mean velocity, maximum velocity, and FVP variables. Within all models, subjects were included as a random intercept and protocol as a random slope to account for the individualized change for each protocol. To identify differences in mean HR and dRPE between training structures, protocol was included as a fixed effect. To assess within protocol changes for the mean and maximum velocity, the first and mean of the last 3 repetitions were included as a categorical fixed effect in addition to protocol. The preresults and postresults for the FVP variables were treated in a similar manner, included within the model as a categorical fixed effect. Residuals of the model were checked for normality through the visual assessment of Q-Q plots. An alpha level was set at 0.05, and Cohen's *d* effect sizes (ESs) with corresponding 90% confidence intervals were calculated by pooling the between subject and residual standard deviation to evaluate between and within protocol differences. Effect size thresholds of

<0.2 trivial, 0.2 small, 0.6 moderate, 1.2 large, and 2.0 very large were reported (13). All data analyses were conducted in RStudio (RStudio: Integrated Development for R. RStudio, PBC, Boston, MA) using the lme4 (3) package.

Results

The TRAD (3.33%; ES = 0.21 ± 0.18 ; $p = 0.015$) and VL5% (3.33%; ES = 0.22 ± 0.18 ; $p = 0.008$) conditions demonstrated *small*, significant within set declines in mean velocity (Figure 2). Alternatively, the RR12 (0.59%; ES = 0.04 ± 0.18 ; $p = 0.99$) and RR4 (1.90%; ES = 0.12 ± 0.18 ; $p = 0.39$) conditions demonstrated *trivial*, nonsignificant reductions in mean velocity (Figure 2). Changes in peak velocity demonstrated similar trends: The TRAD (5.69%; ES = 0.31 ± 0.16 ; $p < 0.001$) and VL5% (4.81%; ES = 0.25 ± 0.16 ; $p < 0.001$) showed *small* and significant decreases, the RR4 condition demonstrated *trivial* and significant decreases (3.01%; ES = 0.16 ± 0.16 ; $p = 0.04$), while the RR12 condition demonstrated *trivial* but nonsignificant changes (1.21%; ES = 0.07 ± 0.15 ; $p = 0.90$) (Figure 3).

No significant between-group differences or effects were observed in dRPE of the legs or breathlessness at any time point (Table 1). In addition, no significant differences in mean heart rate were observed between the RR4 (beats per minute [bpm]: 157.5 ± 9.2), RR12 (bpm: 158.9 ± 10.3), or VL5% (bpm: 157.2 ± 9.2). However, *moderate* significant differences were observed between the TRAD condition (bpm: 149.1 ± 8.6) and the RR4 (ES: 0.69 ± 0.43 ; $p = 0.002$), RR12 (ES: 0.79 ± 0.42 ; $p < 0.001$), and VL5% (ES: 0.65 ± 0.42 ; $p = 0.002$) conditions.

Minimal changes in theoretical maximal velocity (v_0), theoretical maximal relative horizontal force (F_0), and maximal relative power (P_{max}) were observed from pre-to-post across all training structures (Table 2). All findings were deemed *unclear* and nonsignificant apart from *trivial* nonsignificant changes in F_0 in the RR4 condition.

Discussion

The aim of our study was to investigate the effects of traditional, rest redistribution, and velocity-based repeated prescriptive methods on repeated sprint performance, perceived effort, heart rate, and changes in FVP profiles in semiprofessional athletes. Although differences in perception of effort of the legs and breathlessness were not apparent between the 4 prescription methods, each prescriptive method caused different mean and peak running velocities. The RR12 condition allowed the greatest maintenance in mean and peak running velocity, closely followed by the RR4 condition. Alternatively, the TRAD condition demonstrated the greatest reduction in peak velocity, while eliciting a significantly lower mean heart rate when compared with the other methods. These findings suggest that if a practitioner's goal is to expose athletes to the greatest mean and peak velocities during repeated sprint training, they should carefully consider the set configurations that are implemented. Specifically, the use of rest redistribution can maintain sprint performance and internal physiological load without altering perceived effort.

Similar to previous resistance training research investigating the use of rest redistribution (32,33), the RR4 and RR12 conditions enabled

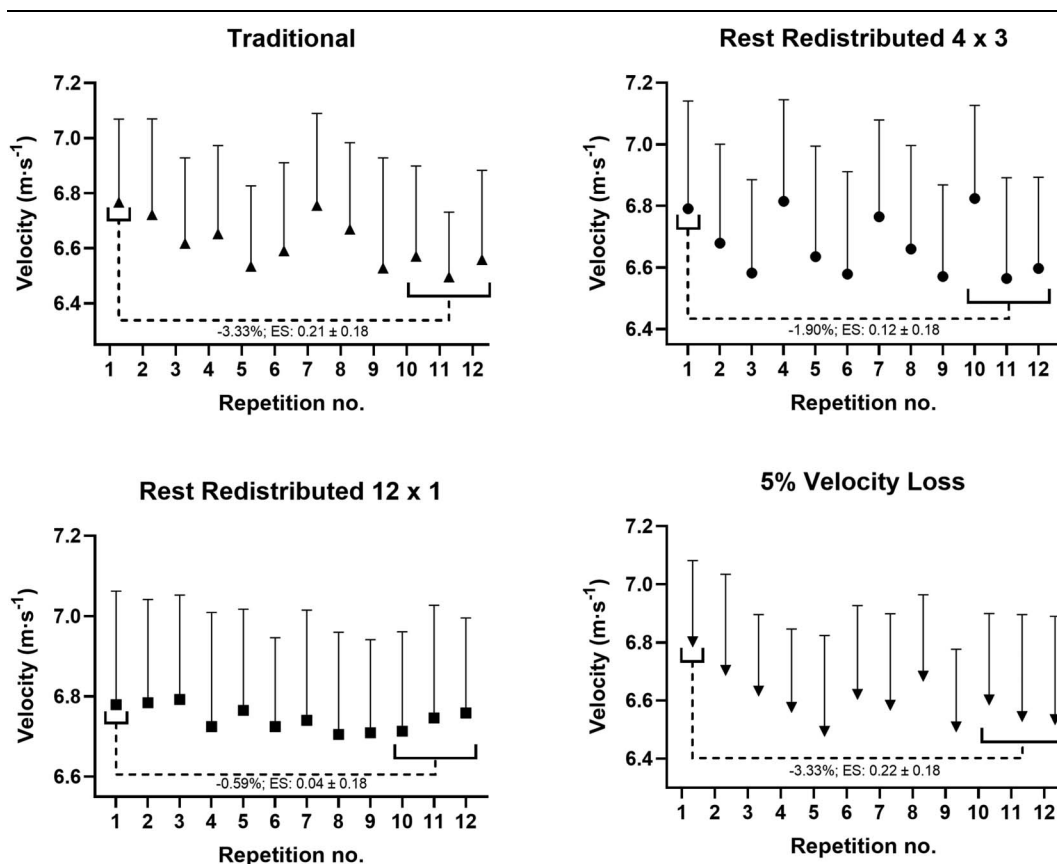


Figure 2. Percentage and ES changes in mean velocity across the 4 different prescription methods. ES = effect size.

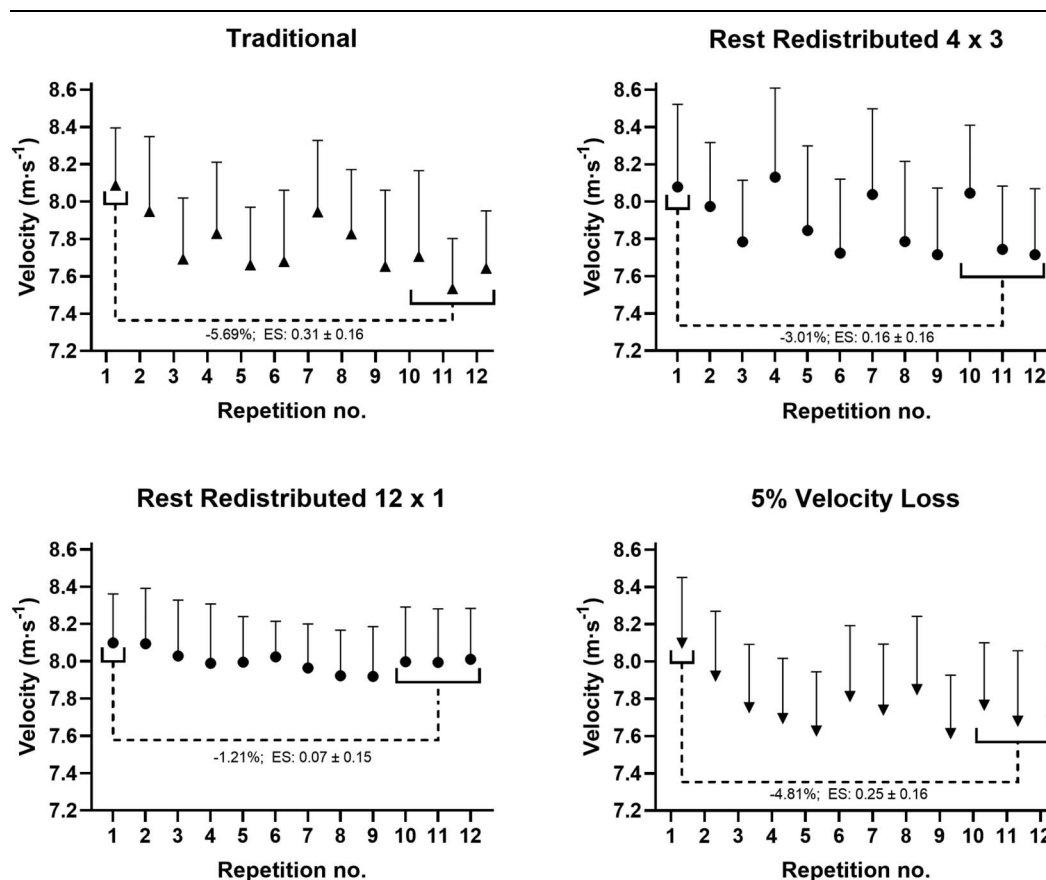


Figure 3. Percentage and ES changes in peak velocity across the 4 different prescription methods. ES = effect size.

the greatest maintenance of mean and peak velocity, with non-significant *small* and *trivial* reductions. **Contrasting this were the TRAD and VL5% conditions that saw significant, *small* decrements in sprint performance.** These findings suggest that to expose athletes to greater training velocities in time-matched conditions, frequent recovery periods that are greater than 20 seconds may be ideal. Alternatively, having a prolonged recovery period (e.g., 4 minutes) between sets can enable the recovery of initial sprint velocity but may reduce the amount of recovery time during sets and hamper subsequent performance (31). Therefore, for practitioners wanting to ensure greater running velocities during repeated sprint training, it is recommended that increased rest (e.g., ~40 seconds) is provided between sprints. In

addition, if short (e.g., 20 seconds) interrepetition rest periods are implemented, a smaller number of repetitions per set may enable greater maintenance of mean and peak sprint performance.

Repeated sprint training is often used as a time efficient method of aerobic conditioning because it can induce substantial elevations in respiration and heart rate (4,5). In this study, there were trivial, nonsignificant differences in average heart rate between the RR12 (158.9 ± 10.3 bpm), RR4 (157.5 ± 9.2 bpm), and VL5% (157.2 ± 9.2 bpm) conditions. However, compared with the other prescriptive methods, the TRAD (149.1 ± 8.6 bpm) condition had a significantly lower average heart rate with a *moderate* effect. This is despite TRAD also showing the greatest reductions in velocity. Furthermore,

Table 1

Differential ratings of perceived exertion for the legs and breathlessness.*

	Third repetition	Sixth repetition	Ninth repetition	12th repetition
Leg dRPE				
TRAD	36.2 ± 14.9	40.0 ± 15.1	43.8 ± 13.5	48.3 ± 14.0
RR4	36.7 ± 8.6	43.3 ± 9.8	47.9 ± 7.8	50.0 ± 10.4
RR12	34.2 ± 9.0	42.5 ± 10.3	46.2 ± 11.3	48.8 ± 10.7
VL5%	36.7 ± 7.8	43.3 ± 7.5	49.6 ± 6.2	49.6 ± 9.6
Breathlessness dRPE				
TRAD	41.7 ± 12.1	49.6 ± 12.9	52.5 ± 14.8	58.3 ± 12.3
RR4x3	45.8 ± 10.0	51.2 ± 10.7	57.9 ± 10.3	61.2 ± 14.0
RR12x1	40.8 ± 10.8	49.6 ± 9.6	56.7 ± 11.5	61.2 ± 13.0
VL5%	45.4 ± 11.4	51.7 ± 12.1	57.9 ± 12.3	60.4 ± 9.6

*dRPE = differential rating of perceived exertion; TRAD = traditional prescription involving 2 sets of 6 repetitions; RR4 = rest redistribution of 4 sets of 3 repetitions; RR12x1 = rest redistribution of 12 sets of 1 repetition; VL5% = velocity loss threshold of 5% from first repetition.

Table 2
Changes in variables derived from the FVP profile from pre-to-post in each session.*

Condition	Pre-	Post-	ES $\pm$ 90% CI	p
F_0 (N $\cdot$ kg $^{-1}$)				
TRAD	6.9 $\pm$ 0.9	7.3 $\pm$ 0.9	0.16 $\pm$ 0.34	0.93
RR4x3	7.5 $\pm$ 1.1	7.4 $\pm$ 1.3	-0.03 $\pm$ 0.36	1.00
RR12x1	7.2 $\pm$ 1.1	7.2 $\pm$ 0.9	-0.01 $\pm$ 0.35	1.00
VL5%	7.2 $\pm$ 0.8	7.4 $\pm$ 1.1	-0.08 $\pm$ 0.35	1.00
v_0 (m $\cdot$ s $^{-1}$)				
TRAD	8.1 $\pm$ 0.6	8.0 $\pm$ 0.4	-0.05 $\pm$ 0.33	1.00
RR4x3	8.0 $\pm$ 0.5	8.2 $\pm$ 0.6	0.10 $\pm$ 0.32	0.94
RR12x1	7.9 $\pm$ 0.5	7.8 $\pm$ 0.5	-0.11 $\pm$ 0.33	0.93
VL5%	8.1 $\pm$ 0.3	8.0 $\pm$ 0.3	-0.10 $\pm$ 0.32	0.96
P_{max} (W $\cdot$ kg $^{-1}$)				
TRAD	13.9 $\pm$ 2.2	14.7 $\pm$ 1.7	0.43 $\pm$ 1.01	0.85
RR4x3	15.0 $\pm$ 2.2	15.0 $\pm$ 2.4	0.01 $\pm$ 1.02	1.00
RR12x1	14.2 $\pm$ 1.6	13.9 $\pm$ 1.6	-0.14 $\pm$ 0.99	1.00
VL5%	15.1 $\pm$ 2.8	14.3 $\pm$ 1.8	-0.37 $\pm$ 1.03	0.92

*ES = effect size; 90% CI = 90% confidence interval; F_0 = theoretical maximal relative horizontal force production; v_0 = theoretical maximal running velocity; P_{max} = maximal relative mechanical power output in the horizontal direction; TRAD = traditional prescription involving 2 sets of 6 repetitions; RR4 = rest redistribution of 4 sets of 3 repetitions; RR12x1 = rest redistribution of 12 sets of 1 repetition; VL5% = velocity loss threshold of 5% from first repetition; FVP = force-velocity-power.

perceptual measures of breathlessness and effort in the legs were nonsignificant with *trivial* differences between all conditions. Together, when completing the TRAD condition, athletes had the greatest reductions in sprinting velocity, the lowest internal load (which is an important consideration for central adaptations), and perceived the session to be equally as taxing as the other conditions. This reduced internal load in the TRAD condition contrasts previous rest redistribution literature in resistance training (13,32,33) and is likely due to the prolonged recovery period between the 2 sets. Therefore, to promote a heightened heart rate response during repeated sprint training, without substantially altering the perception of exertion, it is recommended that rest periods are redistributed more evenly in smaller clusters of repetitions per set or when an athlete's performance drops below a predefined threshold.

Despite significant changes in sprinting performance within the VL5% and TRAD conditions, no significant differences in any condition were observed in F_0 , v_0 , or P_{max} from preintervention to postintervention. This lack of change contrasts previous research, which demonstrated substantial reductions in mechanical outputs during repeated sprint training (14). These outcomes are likely because of the athletes completing a standardized volume, while also being provided a 5-minute recovery period after the 12th sprint which allowed the recovery of PCr and clearance of metabolic by-products for 1 final maximal effort. The measurement of these kinetic and kinematic variables has previously been shown to have satisfactory within-day reliability (29) and suggests that to see a substantial change, greater fatigue may need to be induced and that the recovery of these outputs can be achieved with short periods of recovery.

Although this study is the first to assess and compare the effects of these differing prescription methods on sprint performance, heart rate, perceptual, and kinetic and kinematic alterations during repeated sprint training, it is not without its limitations. First, it should be acknowledged that the interrepetition recovery time can substantially alter training responses and it is likely that performance decrements can be amplified or attenuated depending on the amount of time provided. In this study, interrepetition recovery of 20 seconds was selected because this is commonly used throughout the literature (11,17,25). However, it is acknowledged that the effect of rest

redistribution methods may be altered if different amounts of rest are provided. Second, although extensive piloting occurred so that the velocity loss condition could be as close to time-matched to the other conditions as possible, on average, athletes were provided 30 seconds of additional rest across the entire study when completing the RR12, RR4, and TRAD conditions. In this study, subjects required on average 1.83 ± 0.90 rest periods during the VL5% condition. Owing to the heterogenous nature of repeated sprint capacity in athletes, this slight variation was expected and deemed an amount of time that would not substantially alter a real-world training session. Third, it is important to recognize that velocity-thresholds can be altered so that athletes must perform for longer (i.e., allow for increased velocity loss) or shorter (i.e., reduce the allowed velocity loss) before additional recovery is provided. From piloting, it was decided that for this study, a 5% threshold was the most appropriate because of (a) the average amount of recovery required to enable time-matched protocols, (b) the comparable amount of velocity loss observed within each condition, and (c) the within-session variation in sprint performance that is commonly observed. Finally, it must be acknowledged that while this study investigates the acute effects of these forms of training, the chronic adaptations are still unknown. Consequently, future research should attempt to elucidate the effects of these prescriptive methods on changes in physical qualities.

In conclusion, the findings from our study demonstrate the configuration and prescriptive method used during time-matched repeated sprint training can substantially alter performance outcomes and mean heart rate across the session. However, perception of effort during repeated sprint training and changes in kinetic and kinematic outputs immediately after sprint training completion remain unaffected. Considering this, for practitioners wanting their athletes to maintain sprint performance and promote an elevated heart rate without adversely influencing the perception of effort during repeated sprint training, it is recommended that strategic distribution of recovery is used. In addition, unless increased fatigue within-session fatigue is sought after, smaller repetition numbers within each set, as well as reduced interset rest, may allow for an efficient and effective method of completing repeated sprint training. Finally, future research may wish to further investigate the effects of different velocity loss thresholds during repeated sprint training on mechanical, perceptual, and internal responses.

Practical Applications

Repeated sprint training is a commonly used method of conditioning for athletes and is traditionally prescribed through sets and repetitions. However, large numbers of repetitions with short recovery periods can cause substantial reductions in sprint performance. To better maintain this performance without increasing the training duration or perceived exertion, the strategic manipulation of rest periods can occur. For example, sets can be turned into a rest redistribution format (e.g., separating repetitions into a greater number of smaller sets with more evenly distributed recovery periods). Alternatively, rest can be reallocated so that all repetitions are evenly spaced. Finally, the use of velocity loss thresholds may be a feasible method of prescribing training if athletes have a range of physical qualities (e.g., aerobic capacity), and the decrement of performance beyond a certain point is untenable (e.g., sprint performance declines more than 5% of initial sprint time).

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